

Question	Answer	Marks
10(a)(i)	$L = 4\pi\sigma r^2 T^4$ $3.85 \times 10^{26} = 4\pi \times 5.67 \times 10^{-8} \times r^2 \times 5780^4$	C1
	$r = 6.96 \times 10^8 \text{ m}$	A1

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10(a)(ii)	$F = L / 4\pi d^2$ $= (3.85 \times 10^{26}) / (4\pi \times (1.50 \times 10^{11})^2)$	C1
	$= 1.36 \times 10^3 \text{ W m}^{-2}$	A1
10(a)(iii)	line of same shape showing peak intensity at greater wavelength	B1
	line of same shape showing lower peak intensity	B1
10(b)(i)	5 lines in same pattern shifted to longer wavelengths	B1
10(b)(ii)	$\Delta\lambda / \lambda = v / c$ $\Delta\lambda = (21400 / 300000) \times 656$ $= 46.8 \text{ nm}$	C1
	wavelength = $656 + 46.8$ = 703 nm	A1
10(b)(iii)	(peak) wavelength too high so temperature too low	B1